

D. Power Tower

time limit per test: 4.5 seconds
memory limit per test: 256 megabytes

Priests of the Quetzalcoatl cult want to build a tower to represent a power of their god. Tower is usually made of power-charged rocks. It is built with the help of rare magic by levitating the current top of tower and adding rocks at its bottom. If top, which is built from $k - 1$ rocks, possesses power p and we want to add the rock charged with power w_k then value of power of a new tower will be $\{w_k\}^P$.

Rocks are added from the last to the first. That is for sequence $w_1, ..., w_m$ value of power will be

$$w_1^{\left(w_2^{\left(w_3^{\dots^{w_m}}\right)}\right)}$$

After tower is built, its power may be extremely large. But still priests want to get some information about it, namely they want to know a number called cumulative power which is the true value of power taken modulo m . Priests have n rocks numbered from 1 to n . They ask you to calculate which value of cumulative power will the tower possess if they will build it from rocks numbered $l, l + 1, ..., r$.

Input

First line of input contains two integers n ($1 \leq n \leq 10^5$) and m ($1 \leq m \leq 10^9$).

Second line of input contains n integers w_k ($1 \leq w_k \leq 10^9$) which is the power of rocks that priests have.

Third line of input contains single integer q ($1 \leq q \leq 10^5$) which is amount of queries from priests to you.

k^{th} of next q lines contains two integers l_k and r_k ($1 \leq l_k \leq r_k \leq n$).

Output

Output q integers. k -th of them must be the amount of cumulative power the tower will have if is built from rocks $l_k, l_k + 1, ..., r_k$.

Example

input	Copy
6 1000000000 1 2 2 3 3 3 8 1 1 1 6 2 2 2 3 2 4 4 4 4 5 4 6	
output	Copy
1 1 2 4 256 3 27 597484987	

Note

$3^{27} = 7625597484987$

Codeforces Round 454 (Div. 1, based on Technocup 2018 Elimination Round 4)

Finished

Practice

Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

Last submissions

Submission	Time	Verdict
335835296	Aug/27/2025 19:46	Accepted
335835168	Aug/27/2025 19:45	Time limit exceeded on test 23
335832014	Aug/27/2025 19:13	Accepted
335827887	Aug/27/2025 18:22	Time limit exceeded on test 2
335827437	Aug/27/2025 18:18	Time limit exceeded on test 2
335827331	Aug/27/2025 18:17	Time limit exceeded on test 2
335827163	Aug/27/2025 18:15	Time limit exceeded on test 2
335827016	Aug/27/2025 18:14	Time limit exceeded on test 2
335826764	Aug/27/2025 18:12	Time limit exceeded on test 2
335826669	Aug/27/2025 18:11	Time limit exceeded on test 2

Problem tags

chinese remainder theorem math

number theory *2700